BINOMIAL THEOREM WORKSHEET Y1 AASL

1

1 Write down the first three and last two terms of the following binomial expansions. Do not simplify your answers.

a $(1+2x)^{11}$ **b** $\left(3x+\frac{2}{x}\right)^{15}$ **c** $\left(2x-\frac{3}{x}\right)^{20}$

2 Without simplifying, write down:

a the 6th term of $(2x+5)^{15}$ **b** the 4th term of $\left(x^2+y\right)^9$

c the 10th term of $\left(x-\frac{2}{x}\right)^{17}$ **d** the 9th term of $\left(2x^2-\frac{1}{x}\right)^{21}$.

4 Consider the expansion of $(x+2)^8$.

a Write down the general term of the expansion.

b Find the coefficient of x^5.

5 Consider the expansion of $(x+b)^7$.

a Write down the general term of the expansion.

b Find b given that the coefficient of x^4 is -280.

6 Find the constant term in the expansion of:

a $\left(x+\frac{2}{x^2}\right)^{15}$ **b** $\left(x-\frac{3}{x^2}\right)^9$

7 Find the coefficient of:

a x^{10} in the expansion of $(3+2x^2)^{10}$ **b** x^3 in the expansion of $\left(2x^2-\frac{3}{x}\right)^6$

c x^6y^3 in the expansion of $\left(2x^2-3y\right)^6$ **d** x^{12} in the expansion of $\left(2x^2-\frac{1}{x}\right)^{12}$.

8 Consider the expression $\left(x^2y-2y^2\right)^6$. Find the term in which x and y are raised to the same power.

9 The third term of $(1+x)^n$ is $36x^2$. Find n, and hence find the fourth term.

10 Find a if the coefficient of x^{11} in the expansion of $\left(x^2+\frac{1}{ax}\right)^{10}$ is 15.

11 Find:

a the coefficient of x^4 in the expansion of $(x+4)(x-3)^6$

b the coefficient of x^5 in the expansion of $(x+2)(x^2+1)^8$

c the term containing x^6 in the expansion of $(2-x)(3x+1)^9$.

12 If $(1+kx)^n = 1 - 12x + 60x^2 - \ldots.$, $n \in \mathbb{Z}^+$, find the values of k and n.

9 Find the coefficient of x^5 in the expansion of $(2x+3)(x-2)^6$.

10 Find c given that the expansion $(1+cx)(1+x)^4$ includes the term $22x^3$.

8 Find the coefficient of x^{10} in the expansion of $\left(\frac{3}{x^2} - 4x\right)^{10}$.

9 In the expansion of $\left(3x^2 + \frac{1}{x}\right)^9$, find:

a the coefficient of x^{12} **b** the constant term.

10 The first three terms in the expansion of $(1+kx)^n$, $n \in \mathbb{Z}^+$, are $1 - 4x + \frac{15}{2}x^2$. Find k and n.

11 Find k in the expansion $(m-2n)^{10} = m^{10} - 20m^9n + km^8n^2 - \ldots. + 1024n^{10}$.

12 Find the possible values of q if the constant terms in the expansions of $\left(x^3 + \frac{q}{x^3}\right)^8$ and $\left(x^3 + \frac{q}{x^3}\right)^4$ are equal.

ANSWERS:

1 **a** $1^{11}+\binom{11}{1}(2x)^1+\binom{11}{2}(2x)^2+....+\binom{11}{10}(2x)^{10}+(2x)^{11}$

b $(3x)^{15}+\binom{15}{1}(3x)^{14}\left(\frac{2}{x}\right)^1+\binom{15}{2}(3x)^{13}\left(\frac{2}{x}\right)^2+....$

$....+\binom{15}{14}(3x)^1\left(\frac{2}{x}\right)^{14}+\left(\frac{2}{x}\right)^{15}$

c $(2x)^{20}+\binom{20}{1}(2x)^{19}\left(-\frac{3}{x}\right)^1+\binom{20}{2}(2x)^{18}\left(-\frac{3}{x}\right)^2+....$

$....+\binom{20}{19}(2x)^1\left(-\frac{3}{x}\right)^{19}+\left(-\frac{3}{x}\right)^{20}$

2 **a** $T_6=\binom{15}{5}(2x)^{10}5^5$ **b** $T_4=\binom{9}{3}(x^2)^6y^3$

c $T_{10}=\binom{17}{9}x^8\left(-\frac{2}{x}\right)^9$ **d** $T_9=\binom{21}{8}(2x^2)^{13}\left(-\frac{1}{x}\right)^8$

4 **a** $T_{r+1}=\binom{8}{r}x^{8-r}2^r$ **b** 448

5 **a** $T_{r+1}=\binom{7}{r}x^{7-r}b^r$ **b** $b=-2$

6 **a** $\binom{15}{5}2^5$ **b** $\binom{9}{3}(-3)^3$

7 **a** $\binom{10}{5}3^52^5$ **b** $\binom{6}{3}2^3(-3)^3$ **c** $\binom{6}{3}2^3(-3)^3$

d $\binom{12}{4}2^8(-1)^4$

8 $T_3=\binom{6}{2}(-2)^2x^8y^8$ **9** $n=9,\ T_4=84x^3$

10 $a=2$

11 **a** $\binom{6}{3}(-3)^3+4\binom{6}{2}(-3)^2=0$ **b** $\binom{8}{6}=28$

c $2\binom{9}{3}3^6x^6-\binom{9}{4}3^5x^6=91\,854x^6$

12 $n=6$ and $k=-2$

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9 $8\binom{6}{2}-6\binom{6}{1}=84$ **10** $c=3$

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8 $(-4)^{10}=1\,048\,576$

9 **a** $\binom{9}{2}3^7=78\,732$ **b** $\binom{9}{6}3^3=2268$

10 $k=-\frac{1}{4},\ n=16$ **11** $k=180$ **12** $q=\pm\sqrt{\frac{3}{35}}$